

8255 Programmable Peripheral Interface (PPI) -- MPCA Module 4

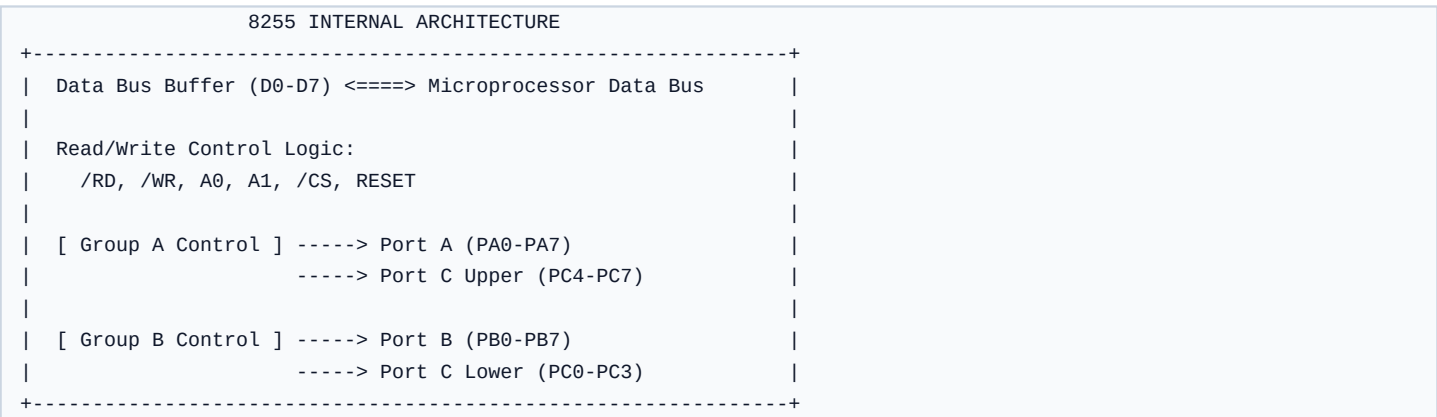
Definition: The Intel 8255A (PPI) is a general-purpose programmable I/O device designed to interface parallel peripheral devices (keyboards, displays, ADCs, printers) to the 8086 microprocessor.

1. Detailed Technical Explanation

1. Internal Block Diagram of 8255

The 8255 features 24 programmable I/O pins divided into three 8-bit ports:

* Po

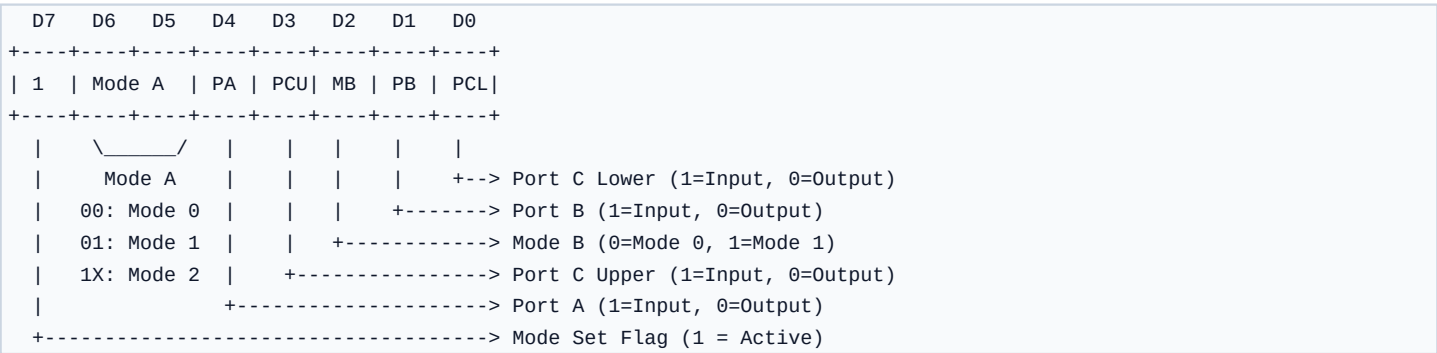


2. 8255 Port Selection Truth Table

A_1	**A_0**	Selected Register / Port
0	0	**Port A**
0	1	**Port B**
1	0	**Port C**
1	1	**Control Word Register (CWR)**
X	X	8255 Deselected (High Impedance)

3. Control Word Register (CWR) Formats

1. I/O Mode Set Control Word (**D_7 = 1**)



2. Operating Modes of 8255:

1. Mode 0 (Basic I/O): Simple input or output without handshake signals. All ports (A, B, C) operate independently.

2. M
INTR).

4. Must-Write Points for Exams

- * 8255 has 24 I/O pins configured into Port A, Port B, and Port C (Upper & Lower).

* D_

5. Quick Recall Flow

8255 PPI -> 24 Pins (Ports A, B, C) -> Port Selection (A1, A0) -> Modes: Mode 0 (Basic), Mode 1 (Strobe), Mode 2 (Bi-dir), BSR Mode